Ludwig Boltzmann, we start with a set of constituents that can mix together in various ways, like atoms and molecules in a fluid. The entropy is then a way of counting the number of ways those constituents can be arranged without changing the macroscopic appearance of the system. Entropy is related to ignorance: high-entropy states are those for which we don't know much about the microscopic details of a system just from knowing its observable features.

Von Neumann entropy, meanwhile, is purely quantum mechanical in nature, and arises from entanglement. Consider a quantum system that is divided into two parts. It could be two electrons, or the quantum fields in two different regions of space. The system as a whole is described by a wave function, as usual. It has some definite quantum state, even if we can only predict measurement outcomes probabilistically. But as long as the two parts are entangled, there is only the one wave function for the whole thing, not a separate wave function for each part. The parts, in other words, are not in definite quantum states of their own.

Von Neumann showed that, for many purposes, the fact that entangled subsystems don't have definite wave functions of their own is analogous to having a wave function, but we just don't know what it is. Quantum subsystems, in other words, closely resemble the classical situation where there are many possible states that look macroscopically the same. And this uncertainty can be quantified into what we now call the *entanglement entropy*. The higher the entropy of a quantum subsystem, the more it's entangled with the outside world.

Think about two qubits, one belonging to Alice and the other to Bob. It might be that they are unentangled, so each qubit has its own wave function, for example, an equal superposition of spin-up and spin-down. In that case, the entanglement entropy of each qubit is zero. Even if we can only predict measurement outcomes probabilistically, each subsystem is still in a definite quantum state.

But imagine that the two qubits are entangled, in an equal superposition of "both qubits are spin-up" and "both qubits are spin-down." Alice's qubit doesn't have its own wave function, because it's entangled with Bob's. Indeed, Bob could perform a measurement of his spin, branching the wave function, so that now there are two copies of Alice, each of whom has a spin in a definite state. But neither copy of Alice knows which state that is; she's in a state of ignorance, where the best she can do is say that there is a